

Md Mobarak Karim

Postdoctoral Associate | Biomedical Optics & Computational Microscopy

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Academic Profile

Postdoctoral Associate in the Richards-Kortum Laboratory, Department of Bioengineering at Rice University, with a Ph.D. in Biomedical Engineering and 7+ years of experience developing optical imaging systems. Research spans optical and computational microscopy, virtual H&E; imaging, array microscopy, Zemax optical design, OCT, one- and two-photon light-sheet fluorescence microscopy, quantitative tissue optics, automated acquisition, and multimodal image analysis.

Research Experience

Postdoctoral Associate - Optical Systems & Computational Microscopy

Aug 2026-Present

Richards-Kortum Laboratory, Department of Bioengineering, Rice University

Houston, TX

Principal Investigator: Dr. Rebecca Richards-Kortum

- Develop optical imaging technologies for AccessPath, including Zemax-based lens and relay design, optical integration, component selection, alignment, and performance characterization.
- Developed an automated unit-microscopy platform integrating motorized XYZ acquisition, synchronized image capture, autofocus, frame averaging, background correction, stitching, and parallel virtual H&E; processing in a unified Python GUI.
- Lead AccessPath APHA-H Phase II optics activities, including optical architecture, system requirements, component evaluation, and phase-mask design, fabrication, and validation for array microscopy.
- Contribute to the PrecisionView optics team through system development, characterization, troubleshooting, and imaging-performance optimization.
- Investigate novel contrast mechanisms and excitation/emission wavelength combinations for virtual H&E; imaging, with emphasis on tissue contrast, background suppression, and pathology-relevant image quality.

Graduate Research Assistant - Multimodal Optical Imaging

Jan 2021-May 2026

Biomedical Optics Laboratory, University of Houston

Houston, TX

Advisor: Dr. Kirill V. Larin

- Designed and integrated a co-registered imaging platform combining 1051-nm swept-source OCT with one-photon and two-photon LSM for simultaneous label-free structural and fluorescence imaging of mouse and zebrafish embryos.
- Achieved ~7-7.6 μm OCT axial resolution, 15 μm lateral resolution, ~100 dB sensitivity, and ~6.9 dB sensitivity roll-off over 4.13 mm; validated resolution, sensitivity, field of view, contrast, and repeatability.
- Engineered beam delivery, polarization management, galvo scanning, scan-lens interfaces, camera/laser synchronization, sample positioning, and environmental control for stable 3D and longitudinal imaging.
- Demonstrated one-photon fluorescence imaging to ~250 μm and two-photon visualization beyond 800 μm in scattering embryonic tissue while OCT provided whole-embryo morphology.
- Developed depth-resolved OCT attenuation algorithms with noise subtraction, confocal correction, and sensitivity roll-off compensation; validated with multilayer TiO₂ phantoms and applied to murine brain development.
- Built MATLAB and Python workflows for OCT reconstruction, multimodal co-registration, morphometry, segmentation, attenuation mapping, 3D visualization, quality control, and standardized image processing.
- Led optical-system integration, biological imaging, quantitative analysis, manuscript preparation, and mentoring of three Ph.D. students and undergraduate researchers.

Graduate Research Assistant - Optical Design & Two-Photon Microscopy

Aug 2018-Dec 2020

3D Biomedical Imaging and Technology Laboratory, GIST

Gwangju, South Korea

- Designed and optimized broadband scan and tube lenses for laser-scanning microscopy using Zemax OpticStudio; evaluated wavefront error, Strehl ratio, aberrations, telecentricity, field of view, and F-theta distortion across 680-1600 nm.
- Developed four optimized scan lenses and one tube lens, reducing F-theta distortion from 13.00 to 1.94 and telecentric distortion from 3.52 to 0.4367.
- Supported two-photon microscope development through optical component evaluation, alignment, testing, measured-performance analysis, and troubleshooting.

Research Expertise

Optical / Computational Microscopy	Automated XYZ microscopy; virtual H&E; array microscopy; phase-mask optics; autofocus; stitching; background correction; frame averaging
Multimodal Imaging	Swept-source OCT; spectral-domain OCT; one- and two-photon LSM; multiphoton; confocal; fluorescence microscopy
Optical Engineering	Zemax OpticStudio; lens/relay design; alignment; 4f relays; galvo scanning; beam delivery; polarization; component qualification
Quantitative Imaging	Attenuation mapping; roll-off/confocal correction; co-registration; segmentation; morphometry; 3D/4D visualization; image-quality metrics
Programming	Python; MATLAB; LabVIEW; C/C++; ImageJ/Fiji; Imaris; Amira; GUI development

Education

Ph.D., Biomedical Engineering

University of Houston

Advisor: Dr. Kirill V. Larin. Dissertation: OCT-LS: A multimodal co-registered optical coherence tomography and light-sheet fluorescence microscopy platform for simultaneous structural and molecular imaging of embryonic development.

Jan 2021-May 2026

Houston, TX

M.S., Biomedical Science and Engineering

Gwangju Institute of Science and Technology

Advisor: Dr. Hyuk-Sang Kwon. Thesis: Analysis of an achromatic doublet as a scan lens for laser-scanning microscopy.

Aug 2018-Dec 2020

Gwangju, South Korea

B.S., Electrical and Electronic Engineering

International Islamic University Chittagong

Aug 2013-Aug 2017

Chittagong, Bangladesh

Peer-Reviewed Journal Articles

- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Yogeshwari S. Ambekar, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, and Kirill V. Larin. "Multimodal optical coherence tomography and two-photon light sheet fluorescence microscopy for embryo imaging." *Journal of Biomedical Optics* 30(6), 060501 (2025). DOI: 10.1117/1.JBO.30.6.060501.
- **Md Mobarak Karim**, Achuth Nair, Manmohan Singh, Maryam Hatami, Salavat R. Aglyamov, and Kirill V. Larin. "Depth-resolved attenuation coefficient quantification during murine embryonic brain development." *Journal of Biophotonics* 18(10), e202500212 (2025). DOI: 10.1002/jbio.202500212.
- Leah A. Lewis, **Md Mobarak Karim**, et al. "Multimodal optical imaging for the assessment of the teratogenic effects of ethanol on zebrafish development." *Journal of Biomedical Optics* 31(6), 066004 (2026). DOI: 10.1117/1.JBO.31.6.066004.
- Mumtaza Mumu, Ayan Das, Talha Bin Emran, Saikat Mitra, Fahadul Islam, Arpita Roy, **Md Mobarak Karim**, et al. "Fucoxanthin: A promising phytochemical on diverse pharmacological targets." *Frontiers in Pharmacology* 13, 929442 (2022). DOI: 10.3389/fphar.2022.929442.

Manuscripts in Preparation or Under Review

- **Md Mobarak Karim**, et al. "Long-term imaging of zebrafish embryonic development with multimodal optical coherence tomography and light-sheet fluorescence microscopy." Manuscript in preparation.
- James L. Klein, **Md Mobarak Karim**, Christian Zevallos-Delgado, Manmohan Singh, Salavat R. Aglyamov, and Kirill V. Larin. "Tartrazine-Based Optical Clearing for Monitoring Embryogenesis with Optical Coherence Tomography." Submitted to *Optics Letters*.

Selected Conference Proceedings

- **Md Mobarak Karim**, Leah A. Lewis, Geethangili Madamanchi, Manmohan Singh, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, Arne C. Lekven, and Kirill V. Larin. "OCT-LS: A co-registered OCT and LSFM platform for simultaneous structural and molecular imaging of embryonic development." *Proceedings of SPIE* 13841 (2026). DOI: 10.1117/12.3080842.
- **Md Mobarak Karim**, Leah A. Lewis, Christian Zevallos-Delgado, Oscar E. Ruiz, Geethangili Madamanchi, Alexander W. Schill, Salavat R. Aglyamov, Arne C. Lekven, and Kirill V. Larin. "Combined structural and molecular zebrafish embryonic imaging on ethanol exposure effects during vasculature development." *Proceedings of SPIE* 13327 (2025). DOI: 10.1117/12.3043312.
- **Md Mobarak Karim**, Ruijiao Sun, Oscar E. Ruiz, Leah A. Lewis, Manmohan Singh, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, and Kirill V. Larin. "Imaging zebrafish embryonic development with multimodal optical coherence tomography and light-sheet fluorescence microscopy." *Proceedings of SPIE* 12834 (2024). DOI: 10.1117/12.3001234.
- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Harshdeep S. Chawla, Yogeshwari S. Ambekar, Alexander W. Schill, David Mayerich, Mary E. Dickinson, and Kirill V. Larin. "Multimodal optical coherence tomography and two-photon selective-plane illumination microscopy for embryonic imaging." *Proceedings of SPIE* 12371 (2023). DOI: 10.1117/12.2649508.
- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Harshdeep S. Chawla, David Mayerich, Mary E. Dickinson, and Kirill V. Larin. "Combined optical coherence tomography and light sheet fluorescence microscopy for embryonic imaging." *Proceedings of SPIE* 11952 (2022). DOI: 10.1117/12.2610091.

Advanced Research Training

- **Multiscale Biophotonics Training Program**, Beckman Laser Institute, University of California, Irvine, 2024 - NIH-supported training in fluorescence lifetime imaging, nonlinear microscopy, spectroscopy, and translational biophotonics.
- **CLIMB Biophotonics Summer School**, Beckman Institute, University of Illinois Urbana-Champaign, 2023 - hands-on OCT/OCM construction, reconstruction, and quantitative imaging.
- **Siegman International School on Lasers**, Poland, 2022 - advanced training in laser physics, ultrafast optics, nonlinear optics, and photonic systems.

Selected Honors and Awards

- 2025 - Cover image featured in Journal of Biomedical Optics
- 2024 - NIH Travel Grant, UCI Multiscale Biophotonics Training Program
- 2022 - Zuegel Family Scholarship, Siegman International School on Lasers
- 2022 - OPTICA Siegman International School on Lasers Grant
- 2022 - SPIE Photonics West Travel Grant
- 2020 - Outstanding Poster Award, Biomedical Research Meeting, South Korea
- 2018-2020 - Korean Government Scholarship for M.S. study at GIST

Mentoring and Professional Service

- Mentored three Ph.D. students and several undergraduate researchers in OCT/LSFM alignment, microscopy operation, image acquisition, MATLAB/Python analysis, experimental design, and scientific presentation.
- Reviewer for biomedical optics and photonics journals; Senior Judge, Texas Science and Engineering Fair - Biomedical Division (2022).
- Executive Member, University of Houston SPIE Student Chapter; member of OPTICA; founder and former executive member, Bangladesh Robotics Foundation.